



Water molecules around the moon

Using the measurements from a NASA's Orbiter, scientists observed water moving around on the bright side of the Moon. <https://digit.in/apr19-5>



World's biggest rocket by NASA

NASA is building the world's biggest rocket, but it won't be launched anytime soon. <https://digit.in/apr19-6>

Lockheed
Martin's
Mars Lander

Seven Minutes of Terror

**Novel strategies
are required for land-
ing heavy payloads
on Mars**

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ars is the next big target for space exploration by humans. Mars is the most hospitable

planet in the Solar System apart from the Earth. The space agencies of the USA, Russia and China (NASA, Roscosmos and CNSA) are all building up the capabilities of pioneering human expeditions to the red planet. Private companies

such as SpaceX, Lockheed Martin and Boeing are also involved in the effort. Establishing a colony on Mars is expected to be a global scale effort, which would require international co-operation as well as support from many private companies. One of the significant problems of going to Mars is landing heavy payloads on the surface. Mars has a strong gravity, 0.38 times that of the Earth, but a very tenuous atmosphere, which is only 0.6 times as thick as the Earth's. The heaviest payload to have successfully landed on is the Curiosity rover, which weighed one metric ton. Future expeditions would require payloads in the range of 5 to 20 tons. A manned mission to Mars would require a capacity to land somewhere between 40 to 80 tons on the surface. There are a lot of complex operations taking place in just a few minutes to execute a successful landing. The Curiosity and InSight missions for example, entered the Martian atmosphere at

high speeds and used a heat shield to keep the instruments inside cool. Then, the parachute popped out, and after a couple of kilometers of descent, the heat shield separated. Then radar systems on board had to figure out when to fire the retro thrusters, which boost the spacecraft away from the surface of the planet. If the thrusters were fired too soon, they would run out of fuel before the spacecraft reached the surface. The parachute then separates, and the thrusters fire at an angle to distance themselves from the parachute and backshell, to prevent the two colliding. The retro thrusters hovered above the ground, while a "sky crane" lowered the missions through bridle cables – the Curiosity rover and the InSight lander to the surface. The payloads touch the ground in a soft landing, after which the descent stage moves away by using the thrusters. Other landing operations have been complicated too, involving retro thrusters, parachutes, and airbags that deploy just before the payload hits the surface. The effectiveness of parachutes decreases with the increasing weight of payloads, as the atmosphere is just not thick enough. The thin atmosphere means using its resistance, or drag for aerobraking manoeuvres is also not very feasible. Retro thrusters are